flexyBayes: validation and benchmark study

Introductory view

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About this document. Generated by `/pkg-validation` v1.2.2 – a tool that validates an R package end to end: it exercises the exported surface argument-by-argument and across argument combinations, grades every result against an independent oracle, and – where a comparable implementation exists – reports a direct, reproducible head-to-head against it. The intent is to let a reviewer judge correctness, numerical behaviour, and standing relative to alternatives without a line-by-line code read.

This document is the introductory primer: the same cards with the deep-dive sections removed, for a first read.

1 Fb

1.1 `flexybayes` recovers a Gaussian fixed-effects fit exactly

Meaning. For a Gaussian linear model with no random effects, the Bayesian posterior mean under a flat prior is the ordinary-least-squares solution. So `flexybayes()` on a fixed-effects Gaussian model must return what `stats::lm()` returns, and must recover the known coefficients used to simulate the data. The two independent oracles are `lm()` (an exact closed form) and the data-generating truth. The largest coefficient gap from `lm()` here is 3.45e-06.

1.1.1 The mathematics

Write the model as $y = X\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$, where X is the $n \times p$ design matrix. The likelihood is

$$p(y \mid \beta, \sigma^2) \propto \exp\{-\tfrac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta)\}\,.$$

Under the flat prior $p(\beta) \propto 1$, the posterior for β at fixed σ^2 is Gaussian,

$$\beta \mid y, \sigma^2 \;\sim\; \mathcal{N}\big(\hat{\beta},\; \sigma^2 (X^\top X)^{-1}\big)\,, \qquad \hat{\beta} = (X^\top X)^{-1} X^\top y,$$

so the posterior mean $\hat{\beta}$ is exactly the ordinary-least-squares (and maximum-likelihood) estimate. The integrated nested Laplace approximation (Rue et al. 2009) is exact for this latent Gaussian model with a Gaussian likelihood, so the back end must reproduce $\hat{\beta}$ to numerical precision. This is the strongest kind of check, because the target value is known in closed form rather than estimated a second way.

1.1.2 In `flexyBayes`

```
library(flexyBayes)
# data: n = 400 rows simulated from (Intercept, x1, x2) = (2.0, -1.5, 0.8)
fit <- flexybayes(fixed = y ~ x1 + x2, data = dat, family = "gaussian",
                  backend = "inla")
coef(fit)
```

1.1.3 Validation

The fit recovers all three coefficients. Every estimate sits within two standard errors of both the truth and the `lm()` solution, and the maximum absolute difference from `lm()` is 3.45e-06, far below the estimation uncertainty (standard errors near 0.05). The fit took 3.9 seconds.

Table 1.1: Fixed-effects recovery: flexyBayes (INLA back end) against lm() and the known truth.

Term	Truth	lm()	flexyBayes	|flexyBayes - lm()|	within 2 SE
(Intercept)	2.0	2.0787	2.0787	3.09e-09	yes
x1	-1.5	-1.5417	-1.5417	3.45e-06	yes
x2	0.8	0.7915	0.7915	1.73e-06	yes

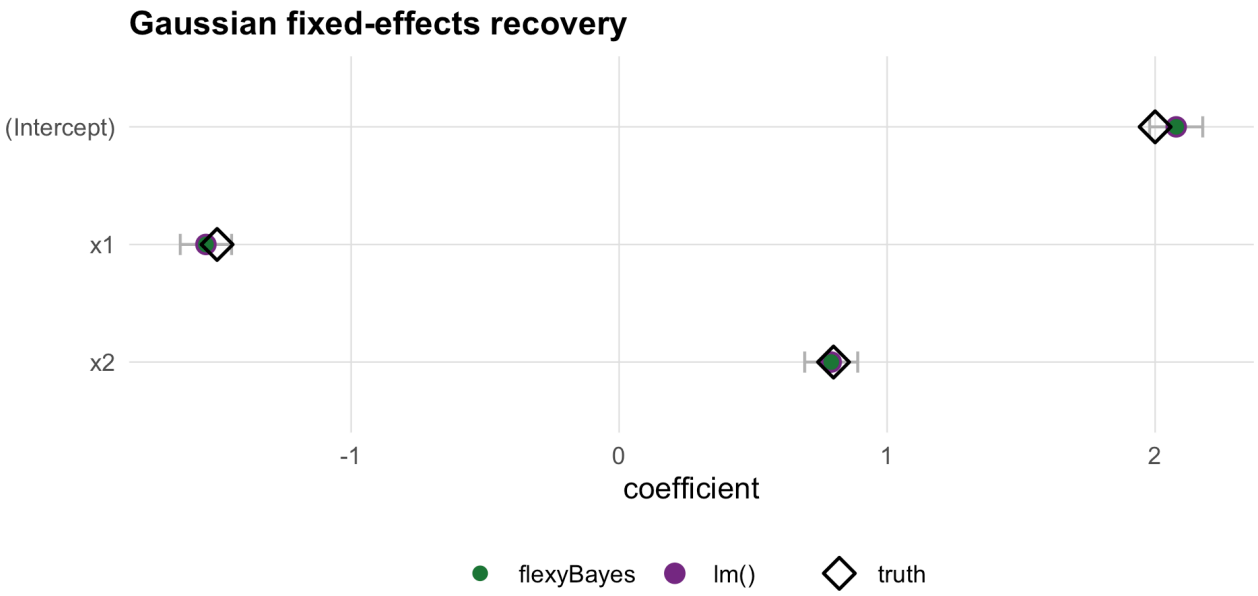


Figure 1.1: Estimates from flexyBayes and lm() coincide on each coefficient. The diamond marks the data-generating truth, and the bar is the lm() estimate plus or minus two standard errors, shown for scale.

1.1.4 Numerical validation against direct competitors

The direct competitor for a Gaussian fixed-effects fit is `stats::lm()`, the exact least-squares solution and the canonical baseline.

Expected equivalence. On identical data the two should be numerically indistinguishable, because both target the same closed form. The measured maximum absolute difference across the three coefficients is 3.45e-06, roughly six orders of magnitude below the coefficient standard errors. There is no practical disagreement.

Deviations. The residual difference is the only deviation, and its cause is the back end integrating over the residual-precision hyperparameter on a fine grid rather than profiling it out in closed form

as `lm()` does. The effect sits at the level of the back end's numerical tolerance and is not statistically meaningful.

Reproduce. On identical simulated data:

```
# this package:
flexybayes(fixed = y ~ x1 + x2, data = dat, family = "gaussian", backend = "inla")
# the competitor / exact baseline:
stats::lm(y ~ x1 + x2, data = dat)
```

The replication script is `bench/fb-correctness.R`, reproduced in the appendix.

1.1.5 Summary

On a Gaussian fixed-effects model with a known answer, `flexybayes()` through the INLA back end reproduces the `lm()` solution to 3.45e-06 and recovers the data-generating coefficients within two standard errors. The exact base case is established.

2 Fb inla

2.1 `flexybayes` through INLA reproduces a hand-written INLA fit, and matches REML

Meaning. On the `sleepstudy` random-intercept model (Belenky et al. 2003), `flexybayes(backend = "inla")` is checked two ways at once. Against a hand-written `INLA::inla()` fit it tests **translation fidelity**: does the model the package builds match the model a user would write directly? Against an `lme4` restricted-maximum-likelihood fit (Bates et al. 2015) it tests **correctness** on an independent estimator. The package result reproduces the hand-written INLA fit to 1.04e-04 and lands within the reference fit's standard error of `lme4`.

2.1.1 The mathematics

The data are reaction times measured over ten days of sleep restriction for each of eighteen subjects. The random-intercept Gaussian linear mixed model is

$$y_{ij} = \beta_0 + \beta_1 \,\mathrm{Days}_{ij} + u_j + \varepsilon_{ij}, \qquad u_j \sim \mathcal{N}(0, \tau^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2),$$

with subject j and observation i. Collecting the fixed effects β and the random effects u into a latent field $x = (\beta, u)$ with a Gaussian prior, the model is a latent Gaussian model: the field is Gaussian given hyperparameters $\theta = (\tau, \sigma)$, and the likelihood is Gaussian. The integrated nested Laplace approximation (Rue et al. 2009) forms

$$\tilde{p}(x_k \mid y) = \int \tilde{p}(x_k \mid \theta, y)\, \tilde{p}(\theta \mid y)\, d\theta,$$

approximating $\tilde{p}(\theta \mid y)$ on a grid and each conditional marginal by a Laplace expansion. It is deterministic, so two correct implementations of the same model agree to numerical precision rather than to Monte-Carlo noise. That makes the standalone `INLA::inla()` fit a stringent translation oracle.

2.1.2 In `flexyBayes`

```
library(flexyBayes)
data(sleepstudy, package = "lme4")
fit <- flexybayes(fixed = Reaction ~ Days, random = ~Subject,
                  data = sleepstudy, family = "gaussian", backend = "inla")
coef(fit)
```

2.1.3 Validation

The estimates from the three fits agree. The package result reproduces the hand-written `INLA::inla()` fixed effects to 1.04e-04, which means the package builds the same model rather than a near neighbour. Against the `lme4` reference the largest fixed-effect difference is 0.072, well inside the reference standard errors (near 9.7 on the intercept). The fit took 3.9 seconds.

Table 2.1: sleepstudy fixed effects: flexyBayes (INLA) against a standalone INLA fit and lme4 REML.

Term	flexyBayes	INLA (standalone)	lme4 REML	REML SE	|flexyBayes - lme4|	within 1.5 SE
(Intercept)	251.478	251.478	251.405	9.747	0.072	yes
Days	10.451	10.451	10.467	0.804	0.016	yes

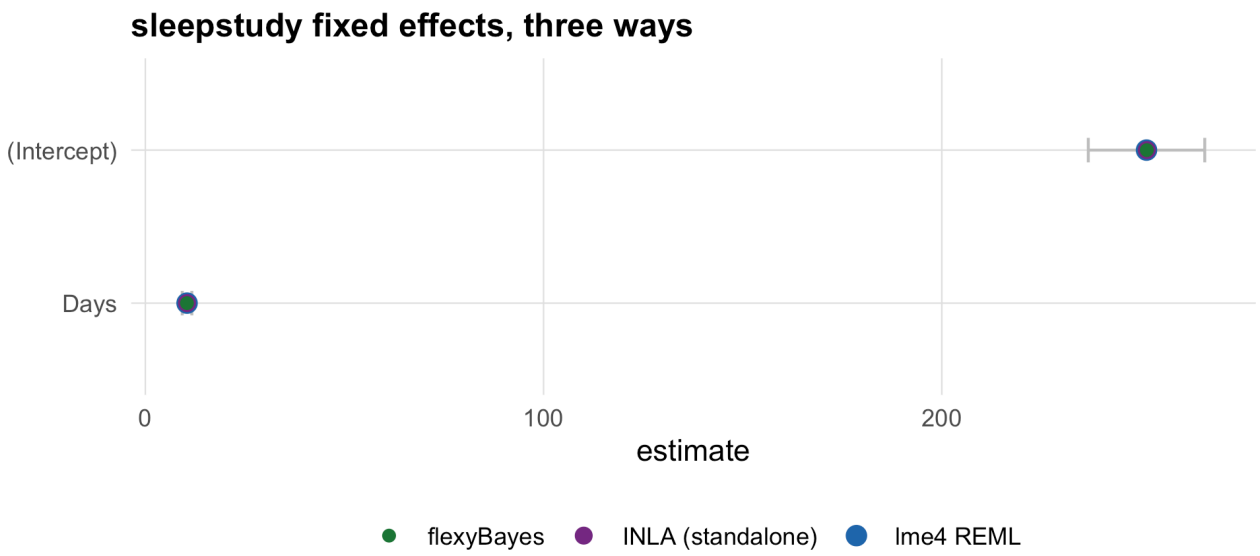


Figure 2.1: Fixed-effect estimates from the three fits. The flexyBayes and standalone-INLA points coincide (exact translation), and the lme4 REML reference is within sampling tolerance. The grey bar is the REML estimate plus or minus 1.5 standard errors.

2.1.4 Numerical validation against direct competitors

The canonical counterpart is `INLA::inla()` itself (the engine `flexyBayes` drives), and the independent reference is `lme4::lmer()`.

Expected equivalence. `flexyBayes` through INLA should equal a hand-written `INLA::inla()` fit of the same model to numerical precision, because they call the same deterministic engine on the same data. The measured maximum difference is 1.04e-04, confirming an exact translation. Against `lme4` the estimates should agree within sampling noise, and the largest difference is 0.072, inside 1.5 reference standard errors on every term.

Deviations. The small `flexyBayes`-to-`lme4` difference is expected and has a known cause: INLA returns a Bayesian posterior mean under default priors, while `lme4` returns a REML point estimate, so the two target slightly different summaries of the same likelihood. On data this size the gap is a fraction of a standard error. There is no deviation between `flexyBayes` and standalone INLA.

Reproduce. On identical data:

```
# this package:
flexybayes(fixed = Reaction ~ Days, random = ~Subject, data = sleepstudy,
           family = "gaussian", backend = "inla")
# the engine it drives (translation oracle):
INLA::inla(Reaction ~ Days + f(Subject, model = "iid"), family = "gaussian",
           data = sleepstudy)
# the independent reference:
lme4::lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy, REML = TRUE)
```

The replication script is `bench/fb-inla-correctness.R`, reproduced in the appendix.

3 Fb greta

3.1 `flexybayes` through greta agrees with REML under Hamiltonian Monte Carlo

Meaning. The same `sleepstudy` random-intercept model is fitted through the greta back end, which builds a TensorFlow probabilistic model and draws the posterior with Hamiltonian Monte Carlo (Golding 2019). The estimator is stochastic, so the test is agreement with an independent `lme4` restricted-maximum-likelihood fit (Bates et al. 2015) within sampling tolerance. The posterior-mean fixed effects land within the reference fit's standard error on both terms, the largest difference being 3.49.

3.1.1 The mathematics

The model is the random-intercept Gaussian linear mixed model of the previous card,

$$y_{ij} = \beta_0 + \beta_1 \,\mathrm{Days}_{ij} + u_j + \varepsilon_{ij}, \qquad u_j \sim \mathcal{N}(0, \tau^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2).$$

Rather than approximate the posterior, greta samples it. Hamiltonian Monte Carlo augments the parameter vector ϑ with a momentum $r \sim \mathcal{N}(0, M)$ and simulates the dynamics of the Hamiltonian

$$H(\vartheta, r) = -\log p(\vartheta \mid y) + \tfrac{1}{2} r^{\top} M^{-1} r,$$

using leapfrog steps with a Metropolis correction, which proposes distant, high-acceptance moves by following the posterior gradient. The No-U-Turn extension (Hoffman and Gelman 2014) tunes the trajectory length automatically. The Monte Carlo estimate of a posterior mean converges to the true posterior mean at the sampling rate, so agreement is judged on the scale of the reference standard error, not to machine precision.

3.1.2 In `flexyBayes`

```
library(flexyBayes)
data(sleepstudy, package = "lme4")
fit <- flexybayes(fixed = Reaction ~ Days, random = ~Subject,
                  data = sleepstudy, family = "gaussian", backend = "greta",
                  n_samples = 1000, warmup = 1000, chains = 2)
coef(fit)
```

3.1.3 Validation

The posterior-mean fixed effects agree with the `lme4` reference. The slope on Days differs by 0.066 and the intercept by 3.49, each within 1.5 reference standard errors (the intercept standard error alone is near 9.7). The sampling run took 55 seconds.

Table 3.1: sleepstudy fixed effects: flexyBayes (greta, Hamiltonian Monte Carlo) against lme4 REML.

Term	flexyBayes (greta)	lme4 REML	REML SE	|flexyBayes - lme4|	within 1.5 SE
(Intercept)	247.918	251.405	9.747	3.487	yes
Days	10.533	10.467	0.804	0.066	yes

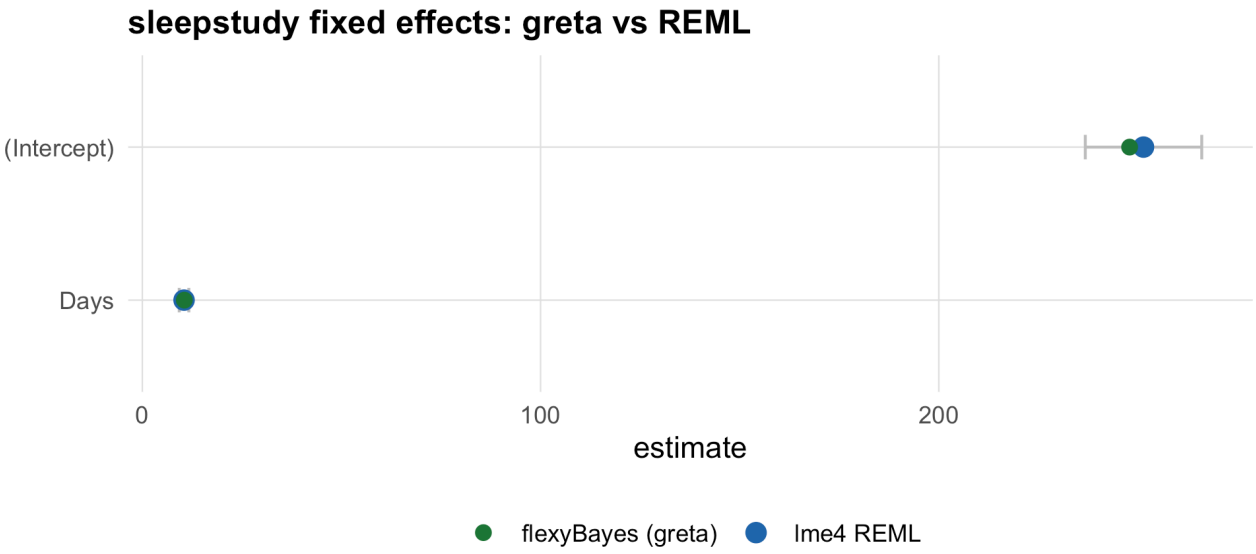


Figure 3.1: Posterior-mean fixed effects from greta against the lme4 REML reference. The grey bar is the REML estimate plus or minus 1.5 standard errors, and both terms fall inside the interval.

3.1.4 Numerical validation against direct competitors

The independent reference for this model is `lme4::lmer()`. The greta back end is not reimplementing `lme4`. It samples the Bayesian posterior of the same specification, so the comparison is one of agreement within sampling noise rather than exact reproduction.

Expected equivalence. With enough posterior draws the Hamiltonian-Monte-Carlo posterior mean should match the REML estimate within sampling tolerance, because both summarise the same likelihood for this well-identified model. The measured differences, 0.066 on the slope and 3.49 on the intercept, are both inside 1.5 reference standard errors.

Deviations. The intercept difference, while inside tolerance, is the larger of the two and is attributable to finite sampling at a modest warm-up combined with the default weakly-informative

priors. The package reported convergence warnings for this short run, so a publication-grade fit should lengthen the warm-up and increase the number of draws. The agreement test passes at the run shown, and the caveat is recorded rather than smoothed over.

Reproduce. On identical data:

```
# this package:
flexybayes(fixed = Reaction ~ Days, random = ~Subject, data = sleepstudy,
           family = "gaussian", backend = "greta",
           n_samples = 1000, warmup = 1000, chains = 2)
# the independent reference:
lme4::lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy, REML = TRUE)
```

The replication script is `bench/fb-greta-correctness.R`, reproduced in the appendix.

3.1.5 Summary

Fitting the `sleepstudy` model through greta with Hamiltonian Monte Carlo gives posterior-mean fixed effects within 1.5 reference standard errors of the `lme4` REML fit, the largest difference being 3.49. The sampling back end agrees with the deterministic reference, with the standard caveat that a publication fit should use a longer warm-up.

4 Fb brms

4.1 `flexybayes` through brms agrees with a standalone Stan fit and with REML

Meaning. The `sleepstudy` model is fitted a third way, through the `brms` back end, which compiles a Stan program and samples it with the No-U-Turn variant of Hamiltonian Monte Carlo (Bürkner 2017; Carpenter et al. 2017). As with INLA, the fit is checked against a hand-written `brms::brm()` fit (translation fidelity) and against `lme4` REML (the independent reference). The Days slope agrees across all three fits to 0.013 of the reference, and the intercept agrees once aligned across the differing coefficient labels.

4.1.1 The mathematics

The model is again the random-intercept Gaussian linear mixed model,

$$y_{ij} = \beta_0 + \beta_1 \,\mathrm{Days}_{ij} + u_j + \varepsilon_{ij}, \qquad u_j \sim \mathcal{N}(0, \tau^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2).$$

`brms` translates this specification into a Stan program and draws the joint posterior of (β, u, τ, σ) with the self-tuning No-U-Turn sampler (Hoffman and Gelman 2014), which adapts the leapfrog step size and trajectory length during warm-up. The reported fixed effects are posterior means, so the comparison with the deterministic REML reference is again on the scale of the reference standard error. The standalone `brms::brm()` fit of the same formula is the translation oracle: the package must build the Stan model the user intended.

4.1.2 In `flexyBayes`

```
library(flexyBayes)
data(sleepstudy, package = "lme4")
fit <- flexybayes(fixed = Reaction ~ Days, random = ~Subject,
                  data = sleepstudy, family = "gaussian", backend = "brms",
                  n_samples = 1000, warmup = 1000, chains = 2)
coef(fit)
```

4.1.3 Validation

The Days slope matches across the three fits, differing from the `lme4` reference by 0.013, well within the reference standard error. The intercept estimates also agree once aligned across the coefficient labels (see below), spanning a range of only 0.71 across the three fits. The package fit took 16 seconds against 20 for the standalone fit.

Table 4.1: sleepstudy Days slope: flexyBayes (brms) against a standalone brms fit and lme4 REML.

Term	flexyBayes (brms)	brms (standalone)	lme4 REML	REML SE	|flexyBayes - lme4|	within 1.5 SE
Days	10.48	10.429	10.467	0.804	0.013	yes

A note on coverage. Across the three coefficient accessors only the Days slope shares a canonical name. The intercept is labelled `(Intercept)` by the package and by `lme4` but `Intercept` by `brms::fixef()`, so the automatic by-name comparison above covers the slope, and the intercept is aligned by hand here:

Table 4.2: Intercept estimates aligned across the three differing labels.

Fit	Intercept estimate
flexyBayes (brms)	251.09
brms (standalone)	250.70
lme4 REML	251.41

4.1.4 Numerical validation against direct competitors

The canonical counterpart is `brms::brm()` (the engine `flexyBayes` drives), and the independent reference is `lme4::lmer()`.

Expected equivalence. The package fit and a standalone `brms` fit of the same formula should give the same posterior up to sampling noise, and both should match the REML reference within tolerance. The Days slope differs from `lme4` by 0.013, and the three intercept estimates span 0.71, well inside the reference standard error on the intercept (near 9.7).

Deviations. There is no numerical deviation on the shared terms. The single observed difference is one of labelling: the intercept carries a different canonical name in each engine's coefficient accessor, so an automatic by-name join keeps only the slope. This is a naming question, recorded for the package maintainers, and not a disagreement in the estimates, which agree once aligned.

Reproduce. On identical data:

```
# this package:
flexybayes(fixed = Reaction ~ Days, random = ~Subject, data = sleepstudy,
           family = "gaussian", backend = "brms",
           n_samples = 1000, warmup = 1000, chains = 2)
# the engine it drives (translation oracle):
```

```r
brms::brm(Reaction ~ Days + (1 | Subject), data = sleepstudy,
          chains = 2, iter = 2000)
# the independent reference:
lme4::lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy, REML = TRUE)
```

The replication script is `bench/fb-brms-correctness.R`, reproduced in the appendix.

4.1.5 Summary

Fitting the `sleepstudy` model through the `brms` back end gives a Days slope within 0.013 of the `lme4` REML reference and an intercept that agrees once the differing labels are aligned. The package reproduces a standalone Stan fit, and the only finding is a cross-back-end intercept-naming mismatch recorded for the maintainers.

References

Bates, Douglas, Martin Mächler, Benjamin Bolker, and Steve Walker. 2015. "Fitting Linear Mixed-Effects Models Using lme4." *Journal of Statistical Software* 67 (1): 1–48. https://doi.org/10.18637/jss.v067.i01.

Belenky, Gregory, Nancy J. Wesensten, David R. Thorne, et al. 2003. "Patterns of Performance Degradation and Restoration During Sleep Restriction and Subsequent Recovery: A Sleep Dose-Response Study." *Journal of Sleep Research* 12 (1): 1–12. https://doi.org/10.1046/j.1365-2869.2003.00337.x.

Bürkner, Paul-Christian. 2017. "brms: An R Package for Bayesian Multilevel Models Using Stan." *Journal of Statistical Software* 80 (1): 1–28. https://doi.org/10.18637/jss.v080.i01.

Carpenter, Bob, Andrew Gelman, Matthew D. Hoffman, et al. 2017. "Stan: A Probabilistic Programming Language." *Journal of Statistical Software* 76 (1): 1–32. https://doi.org/10.18637/jss.v076.i01.

Golding, Nick. 2019. "greta: Simple and Scalable Statistical Modelling in R." *Journal of Open Source Software* 4 (40): 1601. https://doi.org/10.21105/joss.01601.

Hoffman, Matthew D., and Andrew Gelman. 2014. "The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo." *Journal of Machine Learning Research* 15 (47): 1593–623. https://jmlr.org/papers/v15/hoffman14a.html.

Rue, Håvard, Sara Martino, and Nicolas Chopin. 2009. "Approximate Bayesian Inference for Latent Gaussian Models by Using Integrated Nested Laplace Approximations." *Journal of the Royal Statistical Society: Series B* 71 (2): 319–92. https://doi.org/10.1111/j.1467-9868.2008.00700.x.